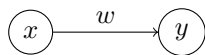


1 Twice.c

1.1 Model



$$y = x \cdot w$$

1.2 Cost

$$\begin{aligned} C &= \frac{1}{n} \sum_{i=1}^n (x_i w - y_i)^2 \\ \frac{dC}{dw} &= \frac{d}{dw} \left(\frac{1}{n} \sum_{i=1}^n (x_i w - y_i)^2 \right) \\ &= \frac{1}{n} \frac{d}{dw} \left(\sum_{i=1}^n (x_i w - y_i)^2 \right) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{d}{dw} \left((x_i w - y_i)^2 \right) \\ &= \frac{1}{n} \sum_{i=1}^n 2(x_i w - y_i) x_i \end{aligned}$$